

Meghana Ganapa

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Available to work immediately |

Summary

Curious data scientist and problem-solver with a Master's in Data Science from the University of Melbourne. Hands-on experience designing LLM-powered systems, NLP pipelines, and data platforms, alongside strong skills in SQL, Python, and analytics engineering. Passionate about building reliable, well-evaluated AI solutions that deliver practical impact.

Education

The University of Melbourne

Feb 2023 – Dec 2024

Masters in Data Science

- Data Science Peer-to-Peer Mentor - provided academic guidance and support on course-specific challenges.

Relevant Experience

Research Intern

Feb 2023 – May 2023

Melbourne Data Analytics Platform – Melbourne, Victoria

- Conducted a comparative analysis of image processing tools (ImageJ and Pixel Annotation Tool) for the purpose of developing a convolutional neural network (CNN) for the analysis of gonadal tissue. Leveraged Python to implement the CNN, resulting in a 15% increase in tissue classification accuracy.

Associate Software Engineer

Sep 2022 – Dec 2022

Qentelli Solutions Private Limited – Hyderabad, Telangana

- Designed and implemented automated workflows utilizing REST APIs, resulting in a 30% reduction in manual effort. Further optimized SQL queries to significantly enhance data retrieval performance.

Software Engineer Intern

Feb 2022 – Aug 2022

Accolite Digital – Hyderabad, Telangana

- Developed and deployed a robust Model-View-Controller (MVC) architecture using Spring Boot and MongoDB. This resulted in a 30% increase in development velocity, a 25% improvement in system reliability, and a significant reduction in production error rates.

Projects

Early Mortality Prediction in Acute Kidney Injury – (Python, SQL)

Mar 2024 – Nov 2024

- Developed a machine-learning pipeline on the MIMIC-IV ICU dataset using SQL, data cleaning, imputation, and LASSO feature selection to identify key predictors of mortality in AKI patients.
- Trained seven ML models and achieved the best performance with XGBoost (AUC: 0.890), outperforming traditional clinical scoring systems.
- Applied SHAP and LIME to provide model interpretability, highlighting critical features such as GCS, blood urea nitrogen, urine output, and age for clinical decision support.

AI-Driven Extraction of Polyp Info from Histology Reports - (Python , Flask , Machine Learning)

Feb 2024 – Nov 2024

- Automated polyp information extraction from histology reports and follow-up colonoscopy recommendations, achieving 83.8% accuracy with LLMs and 69%/67% with CRFs and Naive Bayes for feature extraction.
- Developed a Flask-based user interface enabling medical professionals to easily review extracted data and recommendations.

Melbourne Magic: Interactive Tourist Dashboard for Victorian Travelers -(RShiny, Tableau)

Aug 2023 – Nov 2023

- Integrated multiple public datasets (hospitals, transport, hotels) into an interactive RShiny + Tableau dashboard.
- Enhanced user experience with filters, maps, and recommendations, supporting tourism-related decision-making.

AI-Powered Legal Assistant – (Python, TypeScript, AWS)

May 2025 – June 2025

- Developed and evaluated NLP-based classification models on legal text to support case outcome prediction (80% accuracy).
- Built a secure, AWS-powered platform featuring TypeScript crawlers and dynamic dashboards, allowing legal professionals to analyze case data and accurately gauge the likelihood of success.

Designed & Deployed AWS Data Pipeline for ML-Ready Data (S3, Glue, Redshift)

Aug 2025 – Sept 2025

- Built an end-to-end AWS data pipeline transforming raw CSVs into validated, ML-ready Parquet datasets in S3 using automated Glue ETL workflows.
- Implemented schema inference, data validation, normalization, and partitioning strategies to improve data quality and downstream analytics performance.
- Optimised data storage and access using columnar formats and analytical SQL to enable efficient feature generation and model experimentation.

Technical Skills

Programming & Query Languages: Python, SQL, R, Advanced Microsoft Excel

Data Engineering & Analytics: SQL-based ETL Pipelines, dbt (models, tests, snapshots, macros), Data Modelling (Star Schema), Analytics Engineering, Data Quality & Validation

Cloud Platforms & Warehousing: AWS (S3, Glue, Redshift), Snowflake

Business Intelligence & Visualisation: Power BI (DAX, Power Query), Tableau, Self-Service BI Dashboards, KPI Design

Big Data & Distributed Processing: PySpark (foundational)

Machine Learning & NLP: Scikit-learn, Classification Models, NLP-based Information Extraction, LLM-assisted Analytics

Backend & APIs: Flask, FastAPI, REST APIs

Databases: MySQL, MongoDB

DevOps & Tooling: Docker (basic), Git, Jupyter Notebook, VS Code

Domain Experience: Healthcare & Clinical Data Analytics, Privacy-Aware Data Handling

Achievements

- Published two research papers in Springer: *"Cognitive Baby Care Solutions for Smart Parenting"* and *"Effective Prediction of Brain Tumor Using Machine Learning Algorithms"*.
- Participated in Smart India Hackathon-2018, I-Innovate, and IBM hackathons, showcasing innovative solutions to technical challenges.

Certifications

- AWS Cloud Practitioner
- CCNA